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ADAPTER FOR AN AUTOMATION FIELD DEVICE

Abstract

An adapter for introduction into a two-conductor line comprises an adapter housing; and, arranged in the adapter housing, an adapter electronics that connects a first terminal with a third terminal using a first connecting line and a second terminal with a fourth terminal using a second connecting line, wherein the adapter electronics further includes an overvoltage protection system designed to limit a voltage applied across the third and fourth terminals to a first maximum value in the event of failure; an overcurrent protection system adapted to prevent a rise of the loop current above a maximum electrical current value in the event of failure; and a circuit adapted in the non-failure case to control a voltage applied to the overvoltage protection system, from a minimum input voltage, to a second maximum value lower than the first maximum value.

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Background/Summary

[0001] The invention relates to an adapter for introduction into a two-conductor line between a field device of automation technology and a non-Ex capable supply isolator, in order to make the field device Ex capable, especially Ex-ia capable. The invention also relates to an automation system.

[0002] In automation technology, especially in process automation technology, field devices are often applied, which serve for registering and/or influencing process variables. Serving for registering process variables are sensors, such as, for example, fill level measuring devices, flow measuring devices, pressure-and temperature measuring devices, pH-redox potential measuring devices, conductivity measuring devices, etc., which register the corresponding process variables, fill level, flow, pressure, temperature, pH value, and conductivity. Serving for influencing process variables are actuators, such as, for example, valves or pumps, via which the flow of a liquid in a pipeline section, or the fill level in a container, can be changed. Referred to as field devices are, in principle, all devices, which are applied near to a process and which deliver, or process, process relevant information. In connection with the invention, the terminology, field devices, thus, refers especially also to remote I/Os, radio adapters, and, in general, devices, which are arranged at the field level.

[0003] A large number of such field devices are manufactured and sold by the firm, Endress+Hauser.

[0004] Many field devices are obtainable in so-called 2-conductor versions (also referred to as two-conductor field devices). In such case, energy supply of the field device occurs via the same line pair (two conductor cable), via which also communication occurs.

[0005] Especially in the process industry, however, also in automation technology, physical or technical variables must often be measured, or ascertained, by field devices in regions, in which danger of explosion is present, so-called explosion endangered regions. By suitable measures in the field devices and evaluation systems (such as e.g. voltage-and electrical current limiting), the electrical energy in a signal to be transmitted can be so limited that such signal can under no circumstances (short circuiting, interruptions, thermal effects . . .) trigger an explosion. For this, corresponding protection principles have been established in IEC EN DIN 60079-ff.

[0006] This standard defines, based on the ignition protection types to be applied, structural and circuit measures for the field devices for application in explosion endangered regions. One of these ignition protection types is the ignition protection type “intrinsic safety” (designation Ex-i, IEC EN DIN 60079-11, published June 2012).

[0007] The ignition protection type “intrinsic safety” is based on the principle of limiting electrical current to, and voltage on, an electrical circuit. The energy of the electrical current circuit, which could be able to cause an explosive atmosphere to ignite, is so limited that ignition of the surrounding explosive atmosphere cannot take place, neither by sparks nor by impermissible temperature rise of the electrical components.

[0008] The ignition protection type “intrinsic safety” defines, in such case, three protection levels,

Ex-ia, Ex-ib and Ex-ic. Level a is the highest level, in the case of which two countable failures in their combination do not lead to a malfunction bringing about an ignition (2-failure safety). Level b defines that one countable failure does not lead to a malfunction bringing about an ignition (1-failure safety). Level c correspondingly defines no failure safety, such that in case of a malfunction an ignition can be brought about (0 failure safety).

[0009] In order to achieve protection level Ex-ia, it is necessary in the present state of the art that a so-called Ex-supply isolator be applied, via which an input electrical current and an input voltage applied to the two connection terminals of a field device are limited, e.g. to an input voltage less than 35 V and a maximum power of less than 0.8 W.

[0010] However, there are a large number of already existing installations in automated plants, in the case of which the field devices are operated without an Ex-supply isolator. Such field devices must then satisfy the ignition protection type Ex-m, potting compound encapsulation, or the ignition protection type Ex-d, pressure resistant encapsulation.

[0011] For fulfilling the ignition protection type Ex-m, thus, the parts of the field device, which could ignite an explosive atmosphere by sparks or high temperature, are so embedded in a potting compound that the explosive atmosphere cannot be ignited. This is done by potting the components on all sides with a potting compound resistant against physical-especially electrical, thermal and mechanical—as well as chemical, influences.

[0012] For fulfilling the ignition protection type Ex-d, the parts of the field device, which can ignite an explosive atmosphere, are so arranged in a housing that, when an explosion of an explosive mixture occurs in the inner space of the housing, the housing withstands the explosion pressure safely and a transmitting of the explosion to the explosive atmosphere surrounding the housing is prevented.

[0013] Such field devices require no specially designed Ex-supply isolator, but, instead, can be operated with a conventional supply isolator. When, now, there should be applied in such existing installations a new field device, which is not formed corresponding to the requirements of Ex-d and/or Ex-m, a switching to a specially designed supply isolator is necessary. This switching involves an increased effort for installation of a (new) field device, in order to meet the requirements of Ex-i.

[0014] An object of the invention, thus, is to provide a way in which a field device can meet the protection level Ex-ia without necessitating changing to a specially designed supply isolator in the existing installation.

[0015] The object is achieved according to the invention by the adapter as defined in claim **1** as well as the system of automation technology as defined in claim **12**.

[0016] The adapter of the invention for introduction into a two-conductor line between a field device of automation technology and a non-Ex capable supply isolator, in order to make the field device Ex capable, especially Ex-ia capable, comprises: [0017] an adapter housing having first and second connecting elements for connecting a first two-conductor line, via which the adapter is connectable with the supply isolator, and having third and fourth connecting elements for connecting a second two-conductor line, via which the adapter is connectable with the field device; and, [0018] arranged in the adapter housing, an adapter electronics, which connects the first connecting element with the third connecting element using a first electrical connecting line and the second connecting element with the fourth connecting element using a second electrical connecting line, such that the first connecting line conducts a loop current coming from the supply isolator from the first connecting element to the field device connectable to the third connecting element and the second connecting line conducts the loop current coming from the field device via the fourth connecting element back to the second connecting element at the supply isolator, wherein the adapter electronics further includes an overvoltage protection system, which is designed to limit a voltage applied across the third and fourth connecting elements to a first maximum value in the event of failure, wherein the adapter electronics further includes an overcurrent protection system,

which is adapted to prevent a rise of the loop current above a maximum electrical current value in the event of failure, such that a power issuable to the field device is limited, and wherein the adapter electronics further includes a circuit, which is adapted in the non-failure case to control a voltage applied to the overvoltage protection system, from a minimum input voltage, which lies across the first and second connecting elements, to a second maximum value lower than the first maximum value.

[0019] According to the invention, an adapter is provided, in the case of which from a defined input voltage at the connection terminals, via which the adapter is connected with a supply isolator, the voltage is limited, or controlled, via an overvoltage protection system with the aid of a circuit, especially a semiconductor circuit, to a certain maximum voltage.

[0020] An advantageous embodiment of the adapter of the invention provides that the minimum input voltage applied across the first and second connecting elements is greater than 3 volt, preferably greater than 5 volt, especially preferably greater than 10 volt, quite especially preferably in the range of 10-15 volt.

[0021] Another advantageous embodiment of the adapter of the invention provides that the second maximum value lies in the range from 5 to 25 volt, preferably in the range from 7 volt to 25 volt, especially preferably in the range from 12 to 25 volt, quite especially preferably in the range from 17 to 22 volt.

[0022] Another advantageous embodiment of the adapter of the invention provides that the overvoltage protection system includes at least three Zener diodes, which are connected in parallel with the third and fourth connecting elements, wherein a cathode of the Zener diodes is connected, in each case, with the first connecting line and an anode of the Zener diodes is connected, in each case, with the second connecting line. Especially, the embodiment can provide that the Zener diodes are so selected that a Z voltage lies in the range of 10-20 V, and preferably amounts to about 16 V.

[0023] Another advantageous embodiment of the adapter of the invention provides that the circuit includes a voltage controlled switching element, especially a field effect transistor, for setting a voltage drop, and arranged in the first and/or second connecting line in such a manner that the switching element serves as a voltage divider for a field device connectable to the third and fourth connecting elements.

[0024] Another advantageous embodiment of the adapter of the invention provides that the circuit further comprises an operational amplifier, a voltage divider as well as a feedback loop, wherein the operational amplifier is adapted to control the switching element, wherein the voltage divider is preferably adapted to drive a negative input of the operational amplifier using an intermediate tap as a function of an input voltage lying across the first and second connecting elements and/or the feedback loop preferably serves to lead a voltage drop produced by the switching element back to a positive input of the operational amplifier.

[0025] Another advantageous embodiment of the adapter of the invention provides that the voltage divider is inserted between the first and second connecting lines and, preferably, is embodied as a high ohm voltage divider, such that a transverse current through the voltage divider is not more than 20 μA , preferably not more than 10 μA , quite especially preferably not more than 5 μA .

[0026] An advantageous embodiment of the adapter of the invention provides that the adapter electronics further includes a voltage control unit for the voltage supply of the operational amplifier and is preferably introduced into the first and/or second connecting line in such a manner that the voltage control unit is connected in series with a field device connectable to the third and fourth connecting elements, such that there is no parallel current between the first and second connecting lines. Especially, the embodiment can provide that the voltage control unit comprises a diode, another Zener diode, an additional resistor and/or a voltage stabilizing circuit, especially a reference voltage circuit or a voltage regulator and/or the adapter electronics further includes another Zener diode, which is embodied in such a manner that it is connected in parallel with the

voltage control unit, in order to limit an operating voltage for the operational amplifier to a maximum value and/or the adapter electronics includes a limiting resistor, which is connected in front of the Zener diode to protect the operational amplifier and/or the Zener diode.

[0027] The invention relates further to an automation system comprising a non-Ex-i capable supply isolator, an automation field device, which is embodied in such a manner that it fulfills the requirements of ignition protection type intrinsic safety Ex-i and an adapter as defined in one or more of the above descriptions, wherein the adapter is connected with the non-Ex-i capable supply isolator at the first and second connecting elements by means of a two-conductor line and with the field device at the third and fourth connecting elements by means of an additional two-conductor line.

[0028] An advantageous embodiment of the automation system of the invention and the adapter provides that the field device is further adapted to compensate the transverse current brought about by the voltage divider in the transmitting of measurement-and/or actuating values by means of the loop current.

Description

[0029] The invention will now be explained in greater detail based on the appended drawing, the figures of which show as follows:

[0030] FIG. **1** an arrangement comprising an adapter of the invention, a field device of automation technology and a supply isolator, and

[0031] FIG. **2** an adapter electronics of the invention in detail.

[0032] FIG. **1** shows an arrangement often found in automated plants. In such case, a supply isolator **100** is connected with a field device **300** via a two-conductor line. Via the two-conductor line, measurement-and/or actuating values can be transmitted between the field device **300** and a superordinated unit (not shown), for example, a programmable logic controller (PLC). For example, the measurement-and/or actuating values can be transmitted between the field device and the superordinated unit analogly in the form of a 4-20 mA electrical current signal.

[0033] The supply isolator **100** is a non-Ex capable supply isolator, i.e. it is not permitted/suitable for use in an explosion endangered region. Such non-Ex capable supply isolators are typically found, for example, in already existing installations of automated plants, in which Ex-d devices and/or Ex-m devices are applied, i.e. present.

[0034] The field device **300** can be a field device **300** designed in such a manner that it fulfills the requirements of ignition protection type Ex-m, potting compound encapsulation, or ignition protection type Ex-d, pressure resistant encapsulation. For example, parts of the field device **300**, which can ignite an explosive atmosphere through sparks or high temperature, can be embedded in a potting compound, such that the explosive atmosphere cannot be ignited and the field device is Ex-m capable. Alternatively, parts of the field device **300**, which can ignite an explosive atmosphere, can be so arranged in a housing that when explosion of an explosive mixture occurs in the inner space, the housing safely withstands the explosion pressure and a transmitting of the explosion to explosive atmosphere surrounding the housing is prevented and the field device is Ex-d capable.

[0035] In order to fulfill the requirements of the protection level Ex-ia of the ignition protection type “intrinsic safety”, an adapter of the invention is introduced into the two-conductor line in such a manner that the adapter is arranged between the non-Ex capable supply isolator and the field device **300**. The adapter includes an adapter housing **200** and, arranged in the adapter housing **200**, an adapter electronics **201** with four connection terminals **202**, **203**, **204**, **205** for connecting the two-conductor line.

[0036] FIG. **2** shows an adapter electronics **201** in detail. The adapter electronics **201** includes first

and second connection terminals **202**, **203**, which serve to connect to the first two-conductor line **400**, which comes from the supply isolator, and third and fourth connection terminals **204**, **205**, which serve to connect to the second two-conductor line **500**, which leads to the field device **300**. Furthermore, the adapter electronics **201** includes a first line **206** (in the following also called the plus line), which connects the first connection terminal **202** with the third connection terminal **204**, as well as a second line **207** (in the following also called the minus line), which connects the second connection terminal **203** with the fourth connection terminal **205**.

[0037] In order in the event of failure to limit a voltage on the third and fourth connection terminals **204**, **205**, a number of, preferably three, Zener diodes **D1**, **D2**, **D3** are connected in parallel with the connection terminals. The Zener diodes **D1**, **D2**, **D3** are, in such case, connected in such a manner that a cathode of the Zener diodes is connected with the plus line and an anode with the minus line, such that a Z voltage of the Zener diodes lies across the third and fourth connection terminals **204**, **205** and, thus, defines an output voltage of the adapter for the following field device **300** in failed operation (an Ex-protection measure). The Zener diodes **D1**, **D2**, **D3** are so selected that the Z voltage lies in the range 10-20 V, preferably about 16 V.

[0038] In order in the event of failure to limit an electrical current, which flows from the third connection terminal **204** to the field device **300** and back via the fourth connection terminal **205**, an overcurrent protection system **209**, for example, in the form of a melting fuse, is placed in the plus line. The overcurrent protection system **209** is in combination with the Z voltage of the Zener diodes **D1**, **D2**, **D3** selected in such a manner that a power provided to the field device connected to the third and fourth connection terminals **204**, **205** is limited to a maximum value (Ex power). For example, the overcurrent protection system **209** can be designed to prevent a rise of the electrical current above a maximum electrical current value of 32 mA. Thus, in the example shown in FIG. 2, an Ex power provided on the third and fourth connection terminals is limited to 870 mW ($=32\text{mA} \cdot 16\text{V} \cdot 1.7$), wherein the factor 1.7 is a safety factor per the standard, DIN EN 60079-11.

[0039] In order that the Zener diodes **D1**, **D2**, **D3** are not conducting, or do not become conducting, when an input voltage greater than the Z voltage of the diodes **D1**, **D2**, **D3** lies across the first and second connection terminals **202**, **203**, the adapter electronics **201** of the invention further includes a circuit **600**, which is designed to control, thus to limit, a voltage drop across the Zener diodes **D1**, **D2**, **D3**.

[0040] For this, the circuit **600** comprises a voltage controlled switching element **601** for establishing a voltage drop, wherein the switching element **601** is arranged in such a manner in the first and/or second line **206**, **207** that the switching element **601** serves as a voltage divider for a field device **300** connected to the third and fourth connection terminals **204**, **205**. The switching element **601** can be a transistor, especially a field effect transistor, especially an n-channel, field effect transistor.

[0041] In order appropriately to control the switching element **601**, the circuit **600** further comprises an operational amplifier **602**, a voltage divider **603** as well as a feedback loop **604a**, **604b**. The voltage divider **603** serves to drive a negative input of the operational amplifier **602**. For this, the voltage divider **603** is introduced between the plus-and minus lines **206**, **207** and, preferably, embodied as a high ohm voltage divider. High ohm means, in such case, that a transverse current of not more than 20 μA , especially not more than 10 μA , quite especially not more than 5 μA , flows through the voltage divider. The voltage divider **603** is preferably formed in the ratio 1:10, e.g. a first resistor, which is connected with the plus line, can be 1 megaohm and the second resistor, which is connected to the minus line, 100 kilohm.

[0042] Additionally present in the transverse line of the voltage divider **603** is another Zener diode **D5**, whose cathode is preferably connected with the plus line **206** and whose anode is preferably connected with the minus line **207**. Zener diode **D5** is sized in such a manner that a Z voltage is not greater than the Z voltage of the other Zener diodes **D1**, **D2**, **D3**. For example, the Zener diode **D5** can have a Z voltage of about 13 V. Via the Z voltage of the additional Zener diode, an output

voltage of the adapter, thus an input voltage for the field device, in the failure free operation is defined/determined, as soon as the Z voltage is reached, or exceeded.

[0043] The feedback loop **604a**, **604b** comprises a voltage divider formed of two resistors **604a** and **604b**, wherein the voltage divider is connected in parallel with the switching element **601**, such that a voltage drop produced by the switching element **601** lies across the voltage divider and a positive input of the operational amplifier **602** is connected with a middle tap of the voltage divider. The voltage divider can be formed in the ratio 1:10, in the case of which the resistance with the higher value is connected with the higher voltage. For example, the voltage divider can be formed of a **100** kilohm resistor and a **10** kilohm resistor. As a result of the feedback loop **604a**, **604b**, the operational amplifier **602** is operated in such a manner that such is off on its output, until the voltage drop across the switching element **601** is sufficiently large that the voltage applied via the voltage divider on the positive input of the operational amplifier **602** corresponds to about the voltage on the negative input of the operational amplifier **602**.

[0044] For supplying the operational amplifier **602**, the adapter electronics can have a voltage control unit **605**, which is inserted preferably in the first and/or second line **206**, **207** in such a manner that such is connected in series with a field device **300** connected to the third and fourth connection terminals **204**, **205**. This offers the advantage that there is no current parallel to the third and fourth connection terminals, in order to prevent an influencing of the 4-20 mA electrical current. The voltage control unit **605** can be, for example, a diode, another Zener diode, an additional resistor and/or a reference voltage circuit. The adapter electronics can further comprise another Zener diode **D4**, which is used in such a manner that such is connected in parallel with the voltage control unit **605** and limits an operating voltage of the operational amplifier to a maximum value, e.g. 2 V. Furthermore, the adapter electronics **201** can have a limiting resistor **R1**, which is connected in front of the operational amplifier **602** and/or the additional Zener diode **D4** to protect them.

[0045] An adapter electronics **201** embodied in such a manner assures that in case the supply isolator **100** provides on the first and second connection terminals **202**, **203** a input voltage, which is less than the diode voltage/Z voltage of the Zener diode **D1**, e.g. 12V, the Zener diode **D1** allows no electrical current flow and, thus, the negative input of the operational amplifier **602** is “low” and the positive input “high”, such that the output of the operational amplifier **602** is “high”. This leads to the fact that the switching element **601** is conducting and almost the complete voltage is available via the third and fourth connection terminals **204**, **205** for the field device **300**.

[0046] In the case of a higher input voltage on the first and second connection terminals **202**, **203**, e.g. 30 V, the adapter electronics **201** provides that the other Zener diode **D5** allows an electrical current flow and, thus, the negative input of the operational amplifier **602** lies at a defined voltage value. The voltage value is set via the voltage divider **603**. This voltages depends on the input voltage on the first and second connection terminals **202**, **203**. As soon as the input voltage rises above the diode voltage of the Zener diode **D1**, the input voltage on the operational amplifier **602** also rises proportionally.

[0047] The operational amplifier **602**, thus, provides that a voltage difference between the negative- and positive input is minimized, in that it changes its output voltage and so controls the voltage across the switching element **601**. The voltage, which is controlled with the switching element **601**, is placed via the feedback loop **604a**, **604b** on the positive input of the operational amplifier **602**. This leads to the fact that the operational amplifier **602** in the case of rising input voltage on the first and second connection terminals **202**, **203** operates the switching element **601** in such a manner that it turns off, until the voltage on the positive input of the operational amplifier **602** about equals the voltage on the negative input of the operational amplifier **602**.

LIST OF REFERENCE CHARACTERS

[0048] **100** non-Ex capable supply isolator [0049] **200** adapter housing [0050] **201** adapter electronics [0051] **202** first connecting element [0052] **203** second connecting element [0053] **204**

third connecting element [0054] **205** fourth connecting element [0055] **206** first connecting line [0056] **207** second connecting line [0057] **208** overvoltage protection system [0058] **209** overcurrent protection system [0059] **300** field device of automation technology [0060] **400** first two-conductor line [0061] **500** second two-conductor line [0062] **600** circuit for controlling voltage applied to the overvoltage protection system [0063] **601** switching element, especially a field effect transistor [0064] **602** operational amplifier [0065] **603** voltage divider [0066] **604a**, feedback loop [0067] **604b** feedback loop [0068] **605** voltage control unit [0069] I loop current [0070] **D1** first Zener diode of the overvoltage protection system [0071] **D2** second Zener diode of the overvoltage protection system [0072] **D3** third Zener diode of the overvoltage protection system [0073] **D4** additional diode for limiting operating voltage of the operational amplifier [0074] **R1** limiting resistor connected in front of diode **D4**

Claims

1-12. (canceled)

13. An adapter for introduction into a two-conductor line between a field device of automation technology and a non-Ex capable supply isolator to make the field device Ex capable, the adapter, comprising: an adapter housing having first and second connecting elements for connecting a first two-conductor line via which the adapter is connectable with the supply isolator and having third and fourth connecting elements for connecting a second two-conductor line via which the adapter is connectable with the field device; and an adapter electronics arranged in the adapter housing, wherein the adapter electronics connects the first connecting element with the third connecting element using a first electrical connecting line and the second connecting element with the fourth connecting element using a second electrical connecting line such that the first connecting line conducts a loop current coming from the supply isolator from the first connecting element to the field device connectable to the third connecting element and the second connecting line conducts the loop current coming from the field device via the fourth connecting element back to the second connecting element at the supply isolator, wherein the adapter electronics further includes an overvoltage protection system designed to limit a voltage applied across the third and fourth connecting elements to a first maximum value in the event of failure, wherein the adapter electronics further includes an overcurrent protection system adapted to prevent a rise of the loop current above a maximum electrical current value in the event of failure such that a power issuable to the field device is limited, and wherein the adapter electronics further includes a circuit adapted in the non-failure case to control a voltage applied to the overvoltage protection system from a minimum input voltage, which lies across the first and second connecting elements, to a second maximum value lower than the first maximum value.

14. The adapter as claimed in claim 13, wherein the minimum input voltage applied across the first and second connecting elements is greater than 3 volts.

15. The adapter as claimed in claim 13, wherein the second maximum value lies in the range from 5 to 25 volts.

16. The adapter as claimed in claim 13, wherein the overvoltage protection system includes at least three Zener diodes that are connected in parallel with the third and fourth connecting elements, wherein a cathode of each of the at least three Zener diodes is connected with the first connecting line and an anode of each of the at least three Zener diodes (**D1**, **D2**, **D3**) is connected, in each case, with the second connecting line.

17. The adapter as claimed in claim 15, wherein the at least three Zener diodes are so selected that a Z voltage lies in the range from 10-20 V.

18. The adapter as claimed in claim 13, wherein the circuit includes a voltage controlled switching element for setting a voltage drop and arranged in the first or second connecting line that the switching element serves as a voltage divider for the field device connectable to the third and

fourth connecting elements.

19. The adapter as claimed in claim 18, wherein the circuit further includes an operational amplifier, a voltage divider, as well as a feedback loop, wherein the operational amplifier is adapted to control the voltage controlled switching element, wherein the voltage divider is adapted to drive a negative input of the operational amplifier using an intermediate tap as a function of an input voltage lying across the first and second connecting elements and/or the feedback loop serves to lead a voltage drop produced by the switching element back to a positive input of the operational amplifier.

20. The adapter as claimed in claim 19, wherein the voltage divider is inserted between the first and second connecting lines and is embodied as a high ohm voltage divider such that a transverse current through the voltage divider is not more than 20 μ A.

21. The adapter as claimed in claim 20, wherein the adapter electronics further includes a voltage control unit for the voltage supply of the operational amplifier and introduced into the first and/or second connecting line in such a manner that the voltage control unit is connected in series with a field device connectable to the third and fourth connecting elements such that there is no parallel current between the first and second connecting lines.

22. The adapter as claimed in claim 21, wherein the voltage control unit includes a diode, another Zener diode, an additional resistor and/or a reference voltage circuit and/or the adapter electronics further includes another Zener diode that is embodied in such a manner that it is connected in parallel with the voltage control unit to limit an operating voltage for the operational amplifier to a maximum value and/or the adapter electronics includes a limiting resistor that is connected in front of the other Zener diode to protect the operational amplifier and/or the other Zener diode.

23. An automation system, comprising: a non-Ex-i capable supply isolator; an automation field device that is embodied in such a manner that it fulfills the requirements of ignition protection type intrinsic safety Ex-i; and an adapter, including: an adapter housing having first and second connecting elements for connecting a first two-conductor line via which the adapter is connectable with the supply isolator and having third and fourth connecting elements for connecting a second two-conductor line via which the adapter is connectable with the field device; and an adapter electronics arranged in the adapter housing, wherein the adapter electronics connects the first connecting element with the third connecting element using a first electrical connecting line and the second connecting element with the fourth connecting element using a second electrical connecting line such that the first connecting line conducts a loop current coming from the supply isolator from the first connecting element to the field device connectable to the third connecting element and the second connecting line conducts the loop current coming from the field device via the fourth connecting element back to the second connecting element at the supply isolator, wherein the adapter electronics further includes an overvoltage protection system designed to limit a voltage applied across the third and fourth connecting elements to a first maximum value in the event of failure, wherein the adapter electronics further includes an overcurrent protection system adapted to prevent a rise of the loop current above a maximum electrical current value in the event of failure such that a power issuable to the field device is limited, and wherein the adapter electronics further includes a circuit adapted in the non-failure case to control a voltage applied to the overvoltage protection system from a minimum input voltage, which lies across the first and second connecting elements, to a second maximum value lower than the first maximum value; wherein the adapter is connected with the non-Ex-i capable supply isolator at the first and second connecting elements by a first two-conductor line and with the field device at the third and fourth connecting elements by a second two-conductor line.

24. The automation system as claimed in claim 23, wherein the field device is further adapted to compensate a transverse current brought about by the voltage divider in the transmitting of measurement-and/or actuating values via the loop current.
